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Research on the Phenomenology of HNLs and Possible Signals in Multi-Messenger Observations

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Recently, the IceCube detector has identified the AGN NGC 1068 as a point source of high-energy neutrinos, the γ -ray flux exceeds the prediction of electromagnetic cascade models. This suggests some of the γ ray energy can be transported outside without being attenuated. In this paper, we propose Heavy Neutral Leptons (HNLs) as an explanation for this observation. We first review the origin of the HNL framework theory, calculate the phenomenological behavior of certain processes related to $p - \gamma$ and $p - p$ neutrino production, and use it to explain the γ ray observation of AGN NGC 1068, and several other astronomical observations.

Keywords: HNL; Inverse Seesaw; multi-messenger; NGC 1068

1. Introduction

As advancements are made in astronomical observations, different types of new signals have become observable. Alongside gravitational waves and a wider range of electromagnetic signals, neutrino signals are now observed by IceCube, which has recently spotted AGN NGC 1068 as a point source of neutrino flux around at energies around 1 TeV,¹ and several other experiments, such as KM3NeT/ARCA, while still under construction, have observed a neutrino signal which possibly has energy as high as 220 PeV.²

Furthermore, neutrinos have always been peculiar in Standard Model (SM), because its tiny mass, hence extremely weak coupling to the Higgs field, is considered somewhat unnatural. This has led the neutrino and the origin of its small mass to be considered a gateway to Beyond Standard Model (BSM) physics, with Heavy Neutral Leptons (HNLs) serving as viable dark matter candidates. Numerous terrestrial experiments are dedicated to searching for possible signals coming from BSM theories.³ Although no definite signal has been observed, they do provide a solid constraint on the parameter space of possible BSM theories, for HNL with

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$3 \text{ MeV} < m_N < 19.5 \text{ MeV}$, the mixing angle with muon neutrino is constrained roughly as $|U_{\mu N}|^2 \lesssim 10^{-4}$.⁴

Moreover, the observed γ -ray flux of NGC 1068 exceeds theoretical predictions based on corona cascade models around 10 GeV . This suggests that the hidden core of NGC 1068 has some process to transport more electromagnetic energy out. Axions are one candidate;⁵ another possibility is HNLs, which are discussed in this paper. Before NGC 1068, there are already attempts to combine multi-messenger observations to constrain BSM theories.⁶⁻⁸

This paper is structured as follows. In Section , we outline the basics of HNL theory originating from neutrino mass, and calculate its phenomenology properties, while taking polarization into account. In Section , we investigate the multi-messenger observation results of NGC 1068, try to explain the exceeding γ -ray flux. Conclusions are in Section .

2. ISS Model

To explain the tiny neutrino masses, several mechanisms known as seesaw mechanisms have been proposed. For Type-I Seesaw model, we have

$$m_\nu = \frac{m_D^2}{M_R}, \quad m_N = M_R, \quad |U|^2 = \left(\frac{m_D}{M_R}\right)^2 \quad (1)$$

where m_D is the Dirac mass of neutrino, and M_R is the Majorana mass of the added right-handed neutrino field. For a sufficiently natural m_D and m_R , the mixing angle $|U|^2$ is always too small which makes observation impossible. However, by introducing another fermionic field, the mass matrix becomes

$$(\overline{\nu_L} \quad \overline{\nu_R^c} \quad \overline{S^c}) \begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & M \\ 0 & M & \mu \end{pmatrix} \begin{pmatrix} \nu_L^c \\ \nu_R \\ S \end{pmatrix} \quad (2)$$

The new parameter μ stands for potential lepton number violation, which should be small. The basic properties of the neutrino and a pseudo-Dirac fermion pair, which can be treated as a fermion because of the small μ , are

$$m_\nu = \mu \frac{m_D^2}{M^2}, \quad m_N = M, \quad |U|^2 = \frac{m_\nu}{\mu} \quad (3)$$

3. HNL Production

Such HNL particles interact only through their mixing with SM neutrinos, which implies that processes producing SM neutrinos will also produce HNLs, provided they are kinematically allowed.

3.1. Pion decay

Under AGN conditions, $p-p$ and $p-\gamma$ processes are thought to be the origin of high energy cosmic rays.⁹ Those processes create a large number of pions, whose decays subsequently produce high-energy cosmic rays, including neutrinos and γ -rays.¹⁰

After introducing HNL particles mixing with SM neutrino, charged pions can decay into HNL particles, intermediated by W bosons. By slightly modifying the

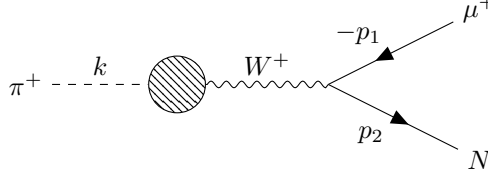


Fig. 1. Feynman diagram of π^+ decaying into μ^+ and massive HNL particle.

process of SM pion decay, we find that the decay width to HNL is

$$\frac{\Gamma_{\pi \rightarrow \mu N}}{\Gamma_{\pi \rightarrow \mu \nu}} = |U_\mu|^2 \times \rho_\pi(m_N) \quad (4)$$

where

$$\rho_\pi(m_N) = \frac{(m_\pi^2(m_\mu^2 + m_N^2) - (m_\mu^2 - m_N^2)^2) \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{m_\mu^2(m_\pi^2 - m_\mu^2)^2} \quad (5)$$

$$\lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx \quad (6)$$

It is useful to determine the helicity of the HNL particle, since it will play a role in the angular dependence when working on the decay of HNL. In the pion rest frame we have

$$\Gamma_{(s)} \sim (m_\mu^2 + m_N^2)m_\pi^2 - (m_\mu^2 - m_N^2)^2 - s(m_\mu^2 - m_N^2) \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)} \quad (7)$$

where s is the helicity of HNL, and $s = +1$ for right-handed polarized HNL particle.

There are no differences in angular dependence for different helicity end states, since pions are unpolarized. After boosting to the lab frame, we must account for the polarization change on the HNL particles, thus for a pion spectrum $J(E_\pi)$, we get

$$Q_N^{(s)}(E_N) = \int_{\lambda^-}^{\lambda^+} J\left(\frac{E_N}{x}\right) |U_\mu|^2 \rho_\pi \times r_{(s)}(x, E_N) \times \frac{dx}{(\lambda^+ - \lambda^-) \cdot x} \quad (8)$$

where

$$\lambda^\pm = \frac{m_\pi^2 - m_\mu^2 + m_N^2 \pm \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{2m_\pi^2} \quad (9)$$

$$\cos \theta_r = \frac{x(m_\pi^2 - m_\mu^2 + m_N^2) - 2m_N^2}{x \sqrt{(1 - \frac{m_N^2}{E_N^2}) \lambda(m_\pi^2, m_\mu^2, m_N^2)}} \quad (10)$$

$$r_{(s)}(x, E_N) = \frac{\Gamma_{(s)} \frac{1 + \cos \theta_r}{2} + \Gamma_{(-s)} \frac{1 - \cos \theta_r}{2}}{\Gamma_{(s)} + \Gamma_{(-s)}} \quad (11)$$

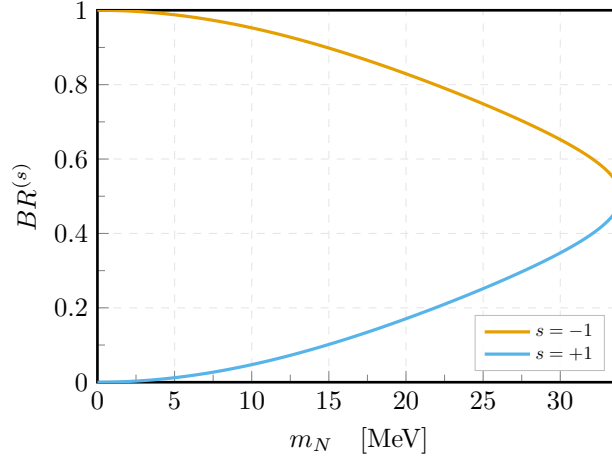
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Fig. 2. Eq. 7 for different s . In pion decay, right-handed HNLs are more likely to be produced close to the phase space boundary.

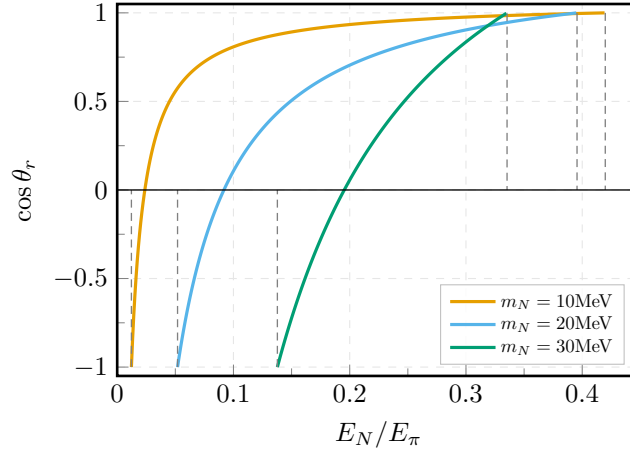


Fig. 3. Eq. 10 visualized with different m_N values. Eq. 10 determines how much helicity flip will happen during Lorentz boost. When m_N is small $\cos \theta_r \approx 1$, combined with Fig. 2, HNLs produced are almost left-handed; as m_N approaches the edge of phase space, the helicity flip is more significant and right-handed HNLs cannot be ignored.

Here we assume $E_\pi \gg m_\pi$ to simplify the boost process. This resembles the SM neutrino case,¹⁰ with modifications arising from HNL mass and helicity separation.

3.2. Muon decay

Muons will decay into HNL particles through a process similar to their decay into ν_μ . We find the decay width of $\mu^- \rightarrow Ne^- \bar{\nu}_e$ has the same form as the process of τ decay which has been studied,¹¹ since this process is symmetric

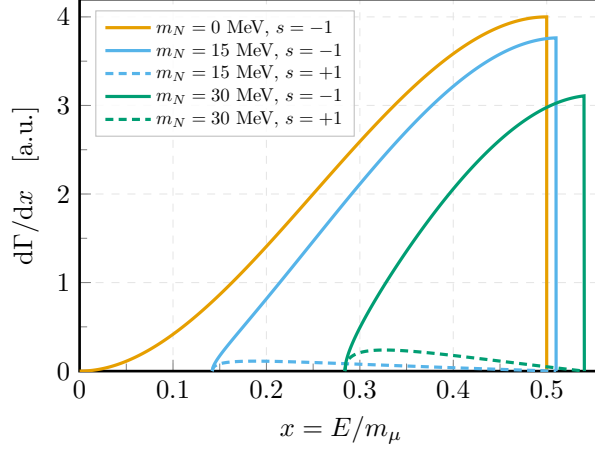


Fig. 4. Energy spectrum of HNL when muons decay into HNLs in muon rest frame. Normalized under $m_N = 0$, which basically resembles SM neutrino, with no right-handed product produced. When m_N increases, shape of the spectrum stays roughly the same, but the max energy increases while the total decay width decreases, yielding a non-negligible fraction of right-handed HNLs.

between μ and τ , which is

$$\frac{\Gamma_{\mu \rightarrow N}}{\Gamma_{\mu \rightarrow \nu_\mu}} = |U_\mu|^2 \times \rho_\mu(m_N) \quad (12)$$

where

$$\rho_\mu(m_N) = -r_N^8 + 8r_N^6 - 24r_N^4 \ln(r_N) - 8r_N^2 + 1, \quad r_N = m_N/m_\mu \quad (13)$$

The helicity-dependent spectrum of the HNL in the muon rest frame is given by

$$\begin{aligned} \frac{d\Gamma}{dE} = \frac{G_F^2}{24\pi^3} |U_\mu|^2 \times |\mathbf{p}| \cdot [3(m_\mu^2 + m_N^2)E - 4m_\mu E^2 - 2m_N^2 m_\mu \\ - s \cdot |\mathbf{p}|(m_N^2 + 3m_\mu^2 - 4m_\mu E)] \end{aligned} \quad (14)$$

Several examples are plotted in Fig. 4.

Although one could retain the muon polarization and compute the angular dependence of the resulting HNL helicity states, we neglect muon polarization here under the assumption that the turbulent magnetic fields in the AGN completely depolarize the muons, in favor of simplifying our calculation.

For a given muon spectrum, we have

$$\begin{aligned} Q_N^{(s)}(E_N) = \int_{\frac{m_N^2}{m_\mu^2}}^1 \frac{dy}{2y} Q_\mu\left(\frac{E_N}{y}\right) |U_\mu|^2 \rho_\mu \cdot \int_{x^-}^{x^+} \frac{dx}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \\ \cdot \left(\frac{1 + \cos \theta_r}{2} g_N^{(s)}(x) + \frac{1 - \cos \theta_r}{2} g_N^{(-s)}(x) \right) \end{aligned} \quad (15)$$

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where

$$x^+ = \frac{m_N^2 + m_\mu^2}{2m_\mu^2}, \quad x^- = \frac{1}{2} \left(y + \frac{1}{y} \cdot \frac{m_N^2}{m_\mu^2} \right) \quad (16)$$

$$\cos \theta_0 = \frac{y - x}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \quad (17)$$

$$\cos \theta_r = \frac{1}{y \sqrt{1 - \frac{m_N^2}{E_N^2}}} \left(\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}} + x \cos \theta_0 \right) \quad (18)$$

where $g_N^{(s)}(x)$ is normalized as

$$\int_{\frac{m_N}{m_\mu}}^{x^+} dx \sum_{s=\pm 1} g_N^{(s)}(x) = 1 \quad (19)$$

4. HNL Decay

Straightforwardly, HNL particles in the ISS model will exclusively decay into other leptons. Possible products are $\nu_\mu e^- e^+$ and $\nu_\mu \nu_l \bar{\nu}_l$. These processes have been well studied;¹¹ their decay widths are given by

$$\Gamma_{N \rightarrow 3\nu} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_\mu|^2 \quad (20)$$

$$\Gamma_{N \rightarrow \nu e \bar{e}} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_\mu|^2 \times C_1^e \quad (21)$$

where

$$C_1^e = \frac{1}{4} (1 - 4 \sin^2 \theta_W + 8 \sin^4 \theta_W) \quad (22)$$

To simplify the problem, we neglect the electron mass. The 3ν final state includes all neutrino flavor combinations. Thus the branching ratio of HNL particle undergoing a visible decay is

$$BR_{N \rightarrow \nu e \bar{e}} = \frac{C_1^e}{1 + C_1^e} \approx 0.11 \quad (23)$$

This results in substantial energy loss, as only approximately one-tenth of the HNL decays are observable. The angular and energy distributions of electrons/positrons are

$$g_e^{(1)}(x, \cos \theta) = \frac{8}{C_1^e} \cdot x^2 \cdot \{[(4 - 16s_W^2 + 64s_W^4)x - (1 - 4s_W^2 + 28s_W^4)] \cos \theta - (4 - 16s_W^2 + 64s_W^4)x + (3 - 12s_W^2 + 36s_W^4)\} \quad (24)$$

$$g_e^{(2)}(x, \cos \theta) = \frac{16}{C_1^e} \cdot x^2 \cdot \{[(6 - 24s_W^2 + 32s_W^4)x - (3 - 12s_W^2 + 14s_W^4)] \cos \theta - (6 - 24s_W^2 + 32s_W^4)x + (3 - 12s_W^2 + 18s_W^4)\} \quad (25)$$

where $x = E_e/m_N$, $s_W = \sin \theta_W$. θ is the angle between the out-going electron and initial HNL spin.

To get the total electron spectrum from a given HNL spectrum, we integrate over the HNL spectrum to obtain

$$Q_e^{(Lab)}(E) = BR_{N \rightarrow \nu e \bar{e}} \cdot \int_E^{+\infty} dE_N Q_N^{(s)}(E_N) \cdot \int_{\frac{E}{2E_N}}^{\frac{1}{2}} \frac{dx}{xE_N} \sum_{i=1,2} g_e^{(i)}(x, s(\frac{E}{xE_N} - 1)) \quad (26)$$

Combining with 24 and 25, the remaining integral is

$$Q_e^{(Lab)}(E) = BR_{N \rightarrow \nu e \bar{e}} \cdot \int_0^1 \frac{dy}{3y} Q_N^{(s)}(\frac{E}{y}) [16y^3 - 27y^2 + 11 - s \cdot (32y^3 - 63y^2 + 36y - 5)] \quad (27)$$

where $y = E/E_N$, and θ_W no longer appears in this expression.

5. HNL in NGC 1068

According to IceCube observations, NGC 1068 has been identified as a neutrino point source, and the dominant channel of neutrino production is charged pion decay.¹² By combining the analytical neutrino spectrum from pion decays¹⁰ with theoretical models of NGC 1068,¹² we adopt the following parameterization as an approximation of the pion spectrum:

$$\Phi_{\pi^\pm}(E) = \Phi_0 \cdot \left(\frac{E}{1 \text{ GeV}} \right)^{-\Gamma} \cdot e^{-(E/E_c)^\beta} \quad (28)$$

where the best fit, which is visualized in Fig. 5, is

$$\Phi_0 = 4.7 \times 10^{-12} \text{ GeV}^{-1} \text{ cm}^{-2} \text{ s}^{-1}, \Gamma = 0.1, E_c = 6.2 \text{ GeV}, \beta = 0.3 \quad (29)$$

By setting $|U_\mu|^2 = 1$ and $m_N = 10 \text{ MeV}$ to 30 MeV , we find that the typical lifetime of HNL particle is in the sub-second region, which means when combined with the actual $|U_\mu|^2 \lesssim 10^{-2}$ the lifetime is longer than that of the muon by 7 to 8 orders of magnitude; hence, it can escape into the outer regions of AGN core. We refer to the parameters of the AGN torus model.¹³ For our approximation we set the magnetic field strength to 139 mG , dust temperature to 1500 K , ambient density to $2.30 \times 10^5 \text{ cm}^{-3}$, bolometric luminosity to $9.55 \times 10^{44} \text{ erg s}^{-1}$, and assume the electron-photon conversion happens at about 0.4 pc away from the AGN core. We use GAMERA to calculate the radiation generated by an input particle spectrum.¹⁴ The result is shown in Fig. 7.

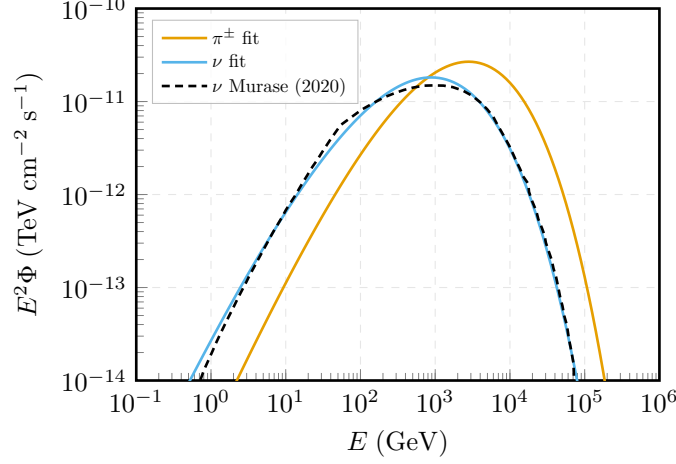


Fig. 5. Parametrization of the pion spectrum in NGC 1068 (Eq. 28) for theoretical prediction.¹²

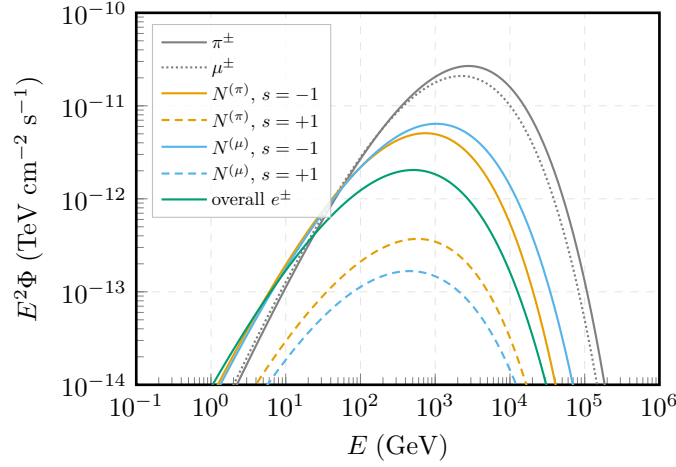


Fig. 6. HNL spectrum from pion and muon decay for NGC 1068, using Eq. 28 as source pion spectrum. The polarization of muon is omitted. HNL mass is set to $m_N = 10$ MeV, and all HNL spectrums are scaled assuming $|U_\mu|^2 = 1$ to better show the physics. For $m_N = 10$ MeV, right-handed neutrino flux is 1 to 2 orders of magnitudes smaller than left-handed one.

6. Conclusions

In this article, we carefully calculated the production and decay processes of HNL particle, assuming it has a mixing angle with muon neutrino. For $\pi^\pm \rightarrow N + \mu^\pm$ and $\mu^\pm \rightarrow N + e^\pm + \bar{\nu}_e$, we calculated the HNL spectrum separated by polarization, and applied Lorentz boost to the lab frame while taking polarization in account. Then we derived the electron/positron spectrum originating from HNL decay with different initial polarizations. We applied this approach to calculate the possible

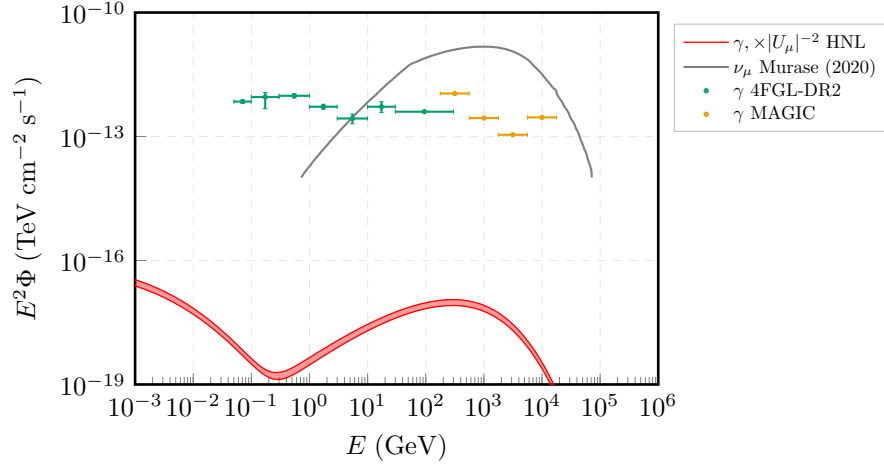


Fig. 7. Possible HNL generated photon spectrum compared with observations. $m_N = 10$ MeV to 30 MeV is shown as a range. Even ignoring the mixing angle, which is smaller than 10^{-4} , the flux is 3 to 4 orders of magnitudes smaller than the observational upper limit, which means it is at least 7 to 8 magnitudes smaller and therefore can be completely ignored in observation.

observables of NGC 1068, assuming that a small mixing angle does exist. While the result suggests it is highly unlikely that this HNL decay route will contribute to any possible observations, the phenomenological framework remains valuable for future studies.

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